Property-Based Testing Lab — Short Report

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Summary

This report summarizes the work completed in each task of the lab-final project (tests implemented/updated under src/test/java/lab/).

Task 1 — Warmup (Task1_WarmupTest.java)

- **Completed:** Read the provided property-based tests and answered the questions in the comment blocks.

- **Changes:** Added concise answers for QUESTION A–F and attached @Report(Reporting.GENERATED) to the palindrome property (prints generated settings and seed). Example output lines were captured in the test run.

Task 2 — countOccurrences (Task2_CountOccurrencesTest.java)

- **Completed:** Implemented five property-based tests for StringUtils.countOccurrences():

1. resultIsNeverNegative — result >= 0
2. zeroOccurrencesWithDisjointCharSets — disjoint char ranges -> count == 0
3. countEqualsRepetitionFactor — text = sub.repeat(n) -> count == n
4. wrappingWithNonMatchingCharsDoesNotChangeCount — adding non-matching prefix/suffix doesn't change count
5. invalidArgumentsThrowException — null/empty inputs raise IllegalArgumentException

- **Notes:** Tests use jqwik constraints and AssertJ assertions. See src/test/java/lab/Task2_CountOccurrencesTest.java

Task 3 — BoundedStack (Task3_BoundedStackTest.java)

- **Completed:** Implemented guided and open properties:

  - lifoOrdering: pushing list then popping yields reverse(list)

  - sizeTracksNumberOfPushes: size() updates with each push and stack becomes full

  - pushPopRoundTrip: push then pop returns same element and size unchanged

  - property4: peek() preserves top element and does not change size

  - property5: pushing to a full stack throws IllegalStateException

- **Notes:** Uses jqwik lists and ranges. See src/test/java/lab/Task3_BoundedStackTest.java

Task 4 — BuggyCalculator (Task4_BuggyCalculatorTest.java)

- **Completed:** Wrote properties to detect intentional bugs and verify behavior:

  - absIsNeverNegative and absProperty2: detect edge-case for Integer.MIN_VALUE where abs fails

  - maxIsAtLeastBothInputs and maxProperty2: detect incorrect max implementation (returns wrong value in some cases)

  - twoIsPrime and isPrimeProperty2: assert isPrime(2) and compare against reference implementation to find bug for small prime

  - clampStaysWithinBounds: verified clamp() behavior (checks result in [min,max] and semantics when value outside/inside bounds)

- Notes: Tests are in src/test/java/lab/Task4_BuggyCalculatorTest.java. Running the tests produces jqwik counterexamples for the buggy methods (use sample in test output to reproduce).

Task 5 — FreeTask (Task5_FreeTaskTest.java)

- **Completed:** Chose Option A (Caesar cipher) and implemented three properties:

  1. shiftBy26IsIdentity — shifting by ±26 returns original text

  2. invertibleByNegatingShift — encrypt(encrypt(text,s),-s) == text

3. nonLettersUnchangedAndCasePreserved — non-letters unchanged, letter case preserved, length preserved

- Implementation includes a `@Provide` generator `printableStrings()` for purposeful input generation.

- File: src/test/java/lab/Task5_FreeTaskTest.java